Yair Schiff

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yair-schiff.github.io

RESEARCH INTERESTS

Generative modeling, Optimal transport, AI for Science, AI for Social Good

EDUCATION

CORNELL UNIVERSITY – DEPARTMENT OF COMPUTER SCIENCE, New York, NY 2022 – Present
PhD candidate in Computer Science
Advisor: Professor Volodymyr Kuleshov

- *Fellowships:* Hal & Inge Marcus PhD Fellowship (2022-2023)
- *Awards:* Cornell Tech Outstanding TA Award (2023)
- *Teaching:* Applied Machine Learning (TA Fall 2022), Deep Learning (TA Spring 2023)

NEW YORK UNIVERSITY – COURANT INSTITUTE OF MATHEMATICAL SCIENCES, New York, NY May 2019
MS in Computer Science, GPA: 3.97/4.00

- *Relevant Coursework:* Advanced Machine Learning, Artificial Intelligence, Computer Vision, Deep Learning, Graphical Processing Units, Machine Learning, Mathematics of Deep Learning, Predictive Analytics

UNIVERSITY OF PENNSYLVANIA – COLLEGE OF ARTS AND SCIENCES, Philadelphia, PA May 2014
BA Summa Cum Laude with Distinction in Economics, GPA: 3.93/4.00

- *Academic Honors:* Phi Beta Kappa, Dean's list 2010-2014

RESEARCH AND PUBLICATIONS

PUBLICATIONS

Learn from Your Mistakes: Self-Correcting Masked Diffusion Models — NeurIPS 2026
Yair Schiff*, *Omer Belhasin**, *Roy Uziel*, *Guanghan Wang*, *Marianne Arriola*, *Gilad Turok*, *Michael Elad*†, *Volodymyr Kuleshov*†
*(*equal contribution)* *(†equal senior authorship)*

d2: Improved Techniques for Training Reasoning Diffusion Language Models — ICML 2026
Guanghan Wang, *Gilad Turok*, ***Yair Schiff***, *Volodymyr Kuleshov*

Encoder-Decoder Diffusion Language Models for Efficient Training and Inference — NeurIPS 2025
*Marianne Arriola**, ***Yair Schiff****, *Hao Phung*, *Aaron Gokaslan*, *Volodymyr Kuleshov*
*(*equal contribution)*

Remasking Discrete Diffusion Models with Inference-Time Scaling — NeurIPS 2025
*Guanghan Wang**, ***Yair Schiff****, *Subham Sekhar Sahoo*, *Volodymyr Kuleshov*
*(*equal contribution)*

Cross-species plant genomes modeling at single nucleotide resolution using a pre-trained DNA language model — PNAS
Jingjing Zhai, *Aaron Gokaslan*, ***Yair Schiff***, *Ana Berthel*, *Zong-Yan Liu*, *Zachary R Miller*, *Armin Scheben*, *Michelle C Stitzer*, *Cinta Romay*, *Edward S. Buckler*, *Volodymyr Kuleshov*

Simple Guidance Mechanisms for Discrete Diffusion Models — ICLR 2025
Yair Schiff, *Subham Sekhar Sahoo*, *Guanghan Wang*, *Hao Phung*, *Sam Boshar*, *Hugo Dalla-torre*, *Bernardo P de Almeida*, *Alexander Rush*, *Thomas Pierrot*, *Volodymyr Kuleshov*

Simple and Effective Masked Diffusion Language Models — NeurIPS 2024
*Subham Sekhar Sahoo, Marianne Arriola, **Yair Schiff**, Aaron Gokaslan, Edgar Marroquin, Justin T Chiu, Alexander Rush, Volodymyr Kuleshov*

Auditing and Generating Synthetic Data with Controllable Trust Trade-offs — IEEE JETCAS
*Brian Belgodere, Pierre Dognin, Adam Ivankay, Igor Melnyk, Youssef Mroueh, Aleksandra Mojsilovic, Jiri Navartil, Apoorva Nitsure, Inkit Padhi, Mattia Rigotti, Jerret Ross, **Yair Schiff**, Radhika Vedpathak, Richard A. Young*

Caduceus: Bi-Directional Equivariant Long-Range DNA Sequence Modeling — ICML 2024
***Yair Schiff**, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, Volodymyr Kuleshov*

DySLIM: Dynamics Stable Learning by Invariant Measure for Chaotic Systems — ICML 2024
***Yair Schiff**, Zhong Yi Wan, Jeffrey B. Parker, Stephan Hoyer, Volodymyr Kuleshov, Fei Sha, Leonardo Zepeda-Núñez*

InfoDiffusion: Representation Learning Using Information Maximizing Diffusion Models — ICML 2023
*Yingheng Wang, **Yair Schiff**, Aaron Gokaslan, Weishen Pan, Fei Wang, Christopher De Sa, Volodymyr Kuleshov*

Semi-Autoregressive Energy Flows: Exploring Likelihood-Free Training of Normalizing Flows — ICML 2023
*Phillip Si, Zeyi Chen, Subham Sekhar Sahoo, **Yair Schiff**, Volodymyr Kuleshov*

Learning with Stochastic Orders — ICLR 2023 (*Notable Top 25% acceptance*)
*Carles Domingo-Enrich, **Yair Schiff**, Youssef Mroueh*

Semi-Parametric Inducing Point Networks and Neural Processes — ICLR 2023
*Richa Rastogi, **Yair Schiff**, Alon Hacohen, Zhaozhi Li, Ian Lee, Yuntian Deng, Mert R. Sabuncu, Volodymyr Kuleshov*

Cloud-Based Real-Time Molecular Screening Platform with MolFormer — ECML PKDD 2022 Demo Track (**alphabetical order equal contribution*)
Brian Belgodere, Vijil Chenthamarakshan*, Payel Das*, Pierre Dognin*, Toby Kurien*, Igor Melnyk*, Youssef Mroueh*, Inkit Padhi*, Mattia Rigotti*, Jarret Ross*, **Yair Schiff***, Richard A. Young**

Optimizing Functionals on the Space of Probabilities with Input Convex Neural Networks — Transactions of Machine Learning Research
*David Alvarez-Melis, **Yair Schiff**, Youssef Mroueh*

Augmenting Molecular Deep Generative Models with Topological Data Analysis Representations — ICASSP 2022 (**equal contribution*)
Yair Schiff, Vijil Chenthamarakshan*, Samuel Hoffman*, Karthikeyan Natesan Ramamurthy*, Payel Das**

Predicting Deep Neural Network Generalization with Perturbation Response Curves — NeurIPS 2021
***Yair Schiff**, Brian Quanz, Payel Das, Pin-Yu Chen*

Tabular Transformers for Modeling Multivariate Time Series — ICASSP 2021
*Inkit Padhi, **Yair Schiff**, Igor Melnyk, Mattia Rigotti, Youssef Mroueh, Pierre Dognin, Jarret Ross, Ravi Nair, Erik Altman*

Image Captioning as an Assistive Technology: Lessons Learned from VizWiz 2020 Challenge — Journal of AI Research
Pierre Dognin, Igor Melnyk*, Youssef Mroueh*, Inkit Padhi*, Mattia Rigotti*, Jarret Ross*, **Yair Schiff***, Richard Young, Brian Belgodere*
*(*alphabetical order equal contribution)*

WORKSHOPS

Advancing DNA Language Models: The Genomics Long-Range Benchmark — AAAI Workshop 2024, ICLR Workshop 2024
*Evan Trop, Chia-Hsiang Kao, Mckinley Polen, **Yair Schiff**, Bernardo P. de Almeida, Aaron Gokaslan, Thomas Pierrot, Volodymyr Kuleshov*

Optimizing Functionals on the Space of Probabilities with Input Convex Neural Networks — NeurIPS Workshop 2021
Spotlight presentation
*David Alvarez-Melis, **Yair Schiff**, Youssef Mroueh*

Gi and Pal Scores: Deep Neural Network Generalization Statistics — ICLR Workshop 2021
***Yair Schiff**, Brian Quanz, Payel Das, Pin-Yu Chen*

Characterizing the Latent Space of Molecular Deep Generative Models with Persistent Homology Metrics — NeurIPS Workshop 2020
Spotlight presentation
***Yair Schiff**, Vijil Chenthamarakshan, Karthikeyan Natesan Ramamurthy, Payel Das*

Alleviating Noisy Data in Image Captioning with Cooperative Distillation — CVPR Workshop 2020
Pierre Dognin, Igor Melnyk*, Youssef Mroueh*, Inkit Padhi*, Mattia Rigotti*, Jarret Ross*, **Yair Schiff**** *(*alphabetical order, equal contribution)*

TALKS AND PRESENTATIONS

Topological Data Analysis and Beyond Workshop — NeurIPS 2020
- Presented spotlight poster "Characterizing the Latent Space of Molecular Deep Generative Models" (video)

VizWiz Grand Challenge Workshop — CVPR 2020
- Presented winning submission to VizWiz Grand Challenge (video)
- Presented "Alleviating Noisy Data in Image Captioning with Cooperative Distillation" (video)

OPEN-SOURCE CONTRIBUTIONS

Caduceus — Github.com
- Authored codebase for long-range DNA sequence modeling using newly proposed Caduceus model.

swirl-dynamics — Github.com
- Added a project for modeling dynamical systems using a regularized objective that aims to preserve systems' invariant measures

SPIN: Semi-Parametric Inducing Point Networks and Neural Process — Github.com
- Implemented Inducing Point Neural Processes and wrote code for training and evaluation

TabFormer: Tabular Transformers for Modeling Multivariate Time Series — Github.com
- Wrote code for training and evaluating GPT-like models on tabular data to generate new, synthetic data that matches the underlying distributions of the real table variables

pytorch-PPUU: Prediction and Policy-learning Under Uncertainty — Github.com
- Added a new dataset on which the self-driving policy could be trained
- Enhanced the self-driving vehicle's policy to enable dynamic lane changes

PROFESSIONAL SERVICES

- ICLR 2027 reviewer
- NeurIPS 2026 reviewer
- ICML 2026 reviewer – Gold Reviewer
- NeurIPS 2025 reviewer – Top Reviewer
- ICML 2025 reviewer
- ICLR 2025 reviewer
- ICML 2024 reviewer
- NeurIPS 2023 reviewer – Top Reviewer

WORK EXPERINCE

INSTADEEP, New York, NY — May 2024 – Nov 2025
PhD Researcher Intern

- Applying generative modeling techniques to genomic sequences
- Investigating control mechanisms for guided sequence generation

GOOGLE, New York, NY — May 2023 – Oct 2023
Student Researcher

- Researched new methods for stabilizing autoregressive rollouts of dynamical system models
- Contributed to internal Google and open-source libraries for modeling dynamical systems

IBM WATSON MACHINE LEARNING, New York, NY — Aug 2019 – Aug 2022
Cognitive Software Developer

- Contributed to continuous development and testing of Watson Machine Learning products
- Facilitated weekly Cloud releases, Cloud Pak for Data platform releases, and the launch of AutoAI feature engineering on relational data, AutoAI Time Series, AutoAI Notebooks, and Federated Learning products
- Published 6 medium.com articles about Watson Machine Learning product releases
- Received Outstanding Technical Achievement Award for work on the release of AutoAI feature engineering on relational data
- Received CrushIT Team Excellence Award as part of the Watson Machine Learning Training team

Research Contributor to IBM Research AI Challenges — Aug 2019 – Aug 2022

- Volunteered to contribute to IBM Research AI challenges, working with the Trusted AI Department
- Member of the first-place winning team in the 2020 VizWiz Grand Challenge: Image Captioning as an Assistive Technology for the Visually Impaired
- Co-authored with the Trusted AI team on several publications in the fields of Generative Modeling, Molecular Discovery, Deep Learning Generalization, and AI for Social Good
- Received two 2021 IBM Research Accomplishments awards for contributions to (1) trustworthy AI generative modeling and (2) deployment of large-scale transformer models on OpenShift environments

SIMON-KUCHER AND PARTNERS, New York, NY — Sept 2014 – Aug 2017
Consultant

- Advised global companies spanning various industries – including internet, media, consumer electronic goods, and chemicals – on areas for better revenue capture
- Synthesized large data sets (e.g., 300 million+ client transactions), customer research (surveys & conjoint studies sent to thousands of respondents), and secondary research to create solutions to client needs

GROUPS AND AFFILIATIONS

GRADS FOR GENDER INCLUSION IN COMPUTING, New York, NY — Present
Member

- Member of a group dedicated to combating harassment, pushing for policy change, and creating supportive spaces

STUDENT-APPLICANT SUPPORT PROGRAM, New York, NY Present
Volunteer

- Provide feedback and support to students applying to graduate programs in Computer Science and related fields

PHD PEER MENTOR PROGRAM, New York, NY Present
Mentor

- Meet with fellow PhD student to provide guidance about graduate school life at Cornell Tech

SKILLS

- *Programming Languages:* C++, Java, Python
- *Deep Learning Frameworks:* PyTorch, PyTorch Lightning
- *Data Tools:* Excel, Stata, Tableau
- *Foreign Languages:* Fluent in Hebrew